



Internship Opportunity: Algo Trading & Quant Research Intern (HFT)

Position: Algo Trading & Quant Research Intern – HFT & Trading Systems

Department: Quantitative Research / Algorithmic Trading / Trading Technology

Location: Mumbai / Indore

Internship Duration: 2–6 Months

PPO Opportunity: Yes – Based on Performance

Role Overview

We are looking for highly analytical and technology-driven Algo Trading & Quant Research Interns who are passionate about High-Frequency Trading (HFT), quantitative strategies, financial markets, and low-latency trading systems.

This is a hands-on role where interns will work on real-world trading strategies, market data analysis, execution systems, and AI-driven quantitative research used in live trading environments.

Interns will collaborate closely with Quant Researchers, Trading Engineers, Product Teams, and Leadership to build scalable and intelligent trading systems.

What You Will Work On

1. Quantitative Research & Strategy Development

- Research and develop quantitative trading strategies.
- Analyze market microstructure and trading behavior.
- Work on statistical and mathematical models for trading signals.
- Backtest trading strategies using historical market data.
- Study arbitrage, momentum, market-making, and volatility-based strategies.

2. Algorithmic & HFT Trading Systems

- Assist in building low-latency trading systems.
- Work on order execution and trade optimization logic.
- Support development of:
 - Strategy execution engines
 - Order management systems (OMS)



- Real-time market data processing
- Risk management systems
- Optimize execution speed and trading performance.

3. AI-Driven Quant & Trading Research

- Use AI tools for:
 - Quantitative research
 - Data analysis
 - Strategy optimization
 - Trading insights
- Experiment with AI-driven trading models and predictive systems.
- Explore ML/AI applications in trading and financial analytics.

4. Data Analytics & Market Research

- Analyze large-scale tick and market datasets.
- Identify patterns, anomalies, and trading opportunities.
- Work on real-time and historical market data analysis.
- Build dashboards and performance tracking systems.

5. Trading Infrastructure & Optimization

- Work with scalable backend and event-driven systems.
- Understand latency optimization and performance tuning.
- Assist in monitoring trading strategy performance and risk exposure.
- Learn infrastructure design for high-performance trading systems.

What You Will Learn

- Real-world HFT and algorithmic trading workflows.
- Quantitative strategy development and backtesting.
- Market microstructure and derivatives trading concepts.
- Low-latency system architecture and execution optimization.



- AI-powered quantitative research techniques.
- How scalable trading infrastructure is built and managed.

Required Technical Skills

Core Skills

- Strong understanding of:
 - Mathematics
 - Statistics
 - Probability
 - Logical reasoning
- Programming knowledge in:
 - Python
 - C++ / Java (Preferred)
- Familiarity with:
 - Data Structures & Algorithms
 - Analytical problem-solving
 - Financial market concepts

Preferred Skills

- Basic understanding of:
 - Stock markets
 - Futures & Options (F&O)
 - Trading systems
 - Quantitative finance
- Exposure to:
 - Backtesting frameworks
 - Market data analysis
 - Time-series analysis
 - SQL
 - Pandas/NumPy



- Competitive programming or quant projects are a plus.
- Active trader/investor is preferred.

AI Knowledge (Mandatory)

Candidates should have basic understanding of:

- Generative AI concepts.
- AI-assisted research and analytics.
- Prompt Engineering basics.
- Machine Learning fundamentals.
- AI-powered trading and prediction systems.

Who Should Apply

- Students from:
 - IITs
 - IISc
 - Top Engineering & Quantitative Institutes
- 3rd/4th year students preferred.
- Candidates passionate about:
 - Quantitative Trading
 - HFT Systems
 - Financial Markets
 - AI-driven trading technologies

What We Are Looking For

- Strong analytical and mathematical mindset.
- Passion for trading systems and quantitative finance.
- Curiosity to solve complex market problems.
- High ownership and execution capability.
- Fast learner with research-oriented thinking.



Internship Benefits

- Opportunity to work on live algorithmic trading systems.
- Exposure to HFT infrastructure and quant research.
- Mentorship from experienced Quant and Trading teams.
- Work on AI-driven trading and analytics systems.
- PPO opportunity for high performers.

Educational Qualification

Students pursuing:

- B.Tech / M.Tech
- Mathematics / Statistics / Computer Science
- AI/ML / Data Science / Quantitative Finance
- Or related fields

How to Apply

Interested candidates can share:

- Resume
- GitHub/LinkedIn/Profile Links
- Quant/Trading/Programming Projects
- Competitive Coding Profiles (Optional)
- Internship Availability Duration